

Homework 1 : Operator approximations

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In this homework, we will explore Koopman/transfer operator approximations of the following 1D dynamical system, $F : [0, 2\pi] \rightarrow [0, 2\pi]$, taken from [Wormell 2024](#):

$$F(x) = 4x - 0.4 \sin 6x + 0.08 \cos 3x \bmod 2\pi. \quad (1)$$

1. In order to approximate the operators, we need samples. We may choose to sample uniformly in $[0, 2\pi]$, but it is more natural to have the samples come from an invariant measure of the system. In this case, convince yourself that the invariant measure has a density with respect to Lebesgue. How do you generate samples from this invariant density? (2 point)
2. Let the samples you generated in the previous part $x_1, \dots, x_m \sim \mu$, where $\rho := d\mu/d\text{Leb}$ is the invariant density from the previous part. Let $\Phi = [\phi_1, \dots, \phi_b]$, be a set of basis functions. Derive, either via a Galerkin formulation or using least squares regression, a matrix approximation, $K \in \mathbb{C}^{b \times b}$ of the Koopman operator, $\mathcal{K}f = f \circ F$. (5 points) Using m samples, the matrix K approximates $P_\Phi \mathcal{K} P_\Phi$, where P_Φ is a projection operator from L^2 to Φ .
3. Choose b to be an even number and $\phi_k = e^{i(k-b/2)}$. Compute your Koopman matrix from the previous part using data Φ at the samples. Plot its eigenvalues, and observe if the elements of the matrix K converge as $m \rightarrow \infty$. Make a convergence plot, and explain your observations. (10 points)
4. Now derive Ulam's method-based approximation of the transfer operator (5 points)
5. Compute the Ulam matrix of size $b \times b$ with b indicator basis functions and plot the eigenvalues. Make a convergence plot for the eigenvalues (10 points)
6. What is the relationship between the eigenvalues and eigenvectors of the Ulam matrix and those of the Koopman matrix as $m \rightarrow \infty$ followed by the limit $b \rightarrow \infty$? (5 points)